

Are You Sitting With Good Posture? Tracking the Position of the Legs via 2-D LiDAR

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Abstract—Desk workers spend long periods sitting with poor posture. This leads to back pain and musculoskeletal disorders. Current solutions to track posture use sensors or cameras that may make sitting uncomfortable and violate privacy. This letter presents a system that detects the leg position when sitting at a desk via 2-D light detection and ranging (LiDAR), and a support vector machine running on a Raspberry Pi 4. The system recognizes six common positions: legs slightly open with knees bent, legs crossed, one leg on the other leg's knee, legs extended forward, backward, and legs parallel with an angle of $\sim 90^\circ$ between upper and lower legs. The results of an experiment involving 30 workers showed an accuracy of 99.2%.

Index Terms—Artificial intelligence, back pain, light detection and ranging (LiDAR), occupational safety and health, privacy preservation, sitting posture.

I. INTRODUCTION

POOR posture causes fatigue and chronic pain as it stresses muscles and joints [1]. For example, sitting with poor posture for long periods may cause musculoskeletal discomfort [2], degrade blood circulation, and lead to neck, back, and shoulder pain, including headache and diminished lung function [3]. Nearly 25% of workers report ongoing back pain worldwide. In the U.S. alone, back and neck pains generate costs for U.S. \$88 billion yearly. Detecting poor posture is thus key.

Some solutions use wearable sensors [4] and intelligent seat pads [5], along with smart chairs based on fuzzy logic [6], processing rules [7], k -nearest neighbors (k -NNs) classifiers [8], and decision trees (DTs) [9]. Other tools use webcams [10], the Microsoft Kinect [11], or real-time location system (RTLS) tags [12]. Although effective, these solutions require sensors on the chair, on the body, or cameras. Body sensors may be annoying, and those on the chair may make sitting uncomfortable. Also, workers may not be willing to work in front of cameras. New devices thus need to track posture but preserve privacy, such as 2-D light detection and ranging (LiDAR), a sensing method that uses a pulsed laser to measure ranges. 2-D

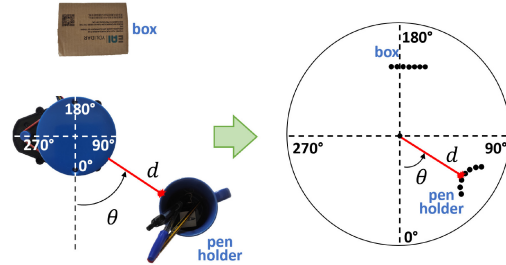


Fig. 1. EAI YDLiDAR X4 sensor (left) and a LiDAR scan (right). The pulse emitted at rotation angle θ (red segment) hits the pen holder at distance d . Black dots in the scan outline the box and pen holder.

LiDAR has recently been used for human activity recognition (HAR) and fall detection [13], and for multiperson HAR [14].

This letter presents a system that uses 2-D LiDAR to track the position of the legs when sitting at a desk. Legs were considered as their incorrect position causes spinal lateral flexion, forward head posture [15], and increased blood pressure [16]. To the best of our knowledge, this is the first work that has used 2-D LiDAR for sitting posture analysis.

The system uses a 2-D LiDAR sensor connected to a Raspberry Pi 4 where a support vector machine (SVM) determines the leg position. The SVM was trained using data collected by 30 office workers, achieving an accuracy of 99.2%.

This letter is organized as follows: Section II overviews the system; Section III describes the data collection; Section IV presents the method; Section V discusses the results; and Section VI draws the conclusions.

II. SYSTEM OVERVIEW

A. 2-D Light Detection and Ranging

A 2-D LiDAR sensor sends laser pulses as it rotates, and detects those that return back after hitting an object in the sensor range (see Fig. 1). For each pulse returned, the sensor saves an *angle-distance pair* (θ, d), where θ is the angle formed by the pulse and a zero-degree reference point, whereas d is the distance of the hit object from the sensor. Nonreturning pulses are assigned $d = 0$. Angle-distance pairs recorded during a 360° rotation form a *LiDAR scan*.

B. System Architecture

An EAI YDLiDAR X4 was stacked on and connected to a Raspberry Pi 4 via a Micro USB port. Fig. 2 shows the device on a tripod. A Raspberry Pi Camera Module 2 was connected to the Raspberry's 15-pin MIPI camera serial interface (CSI-2) to record videos of the legs during data collection. Participants

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

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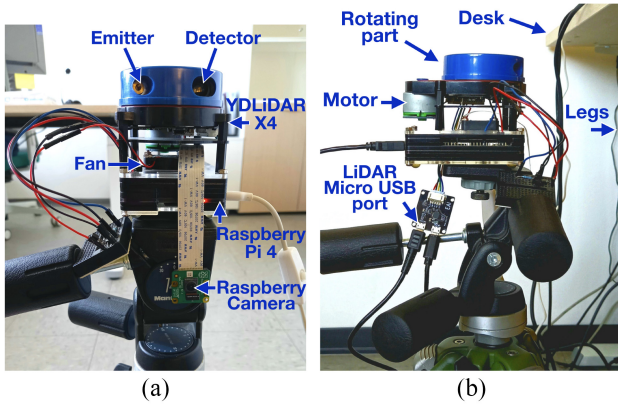


Fig. 2. (a) Front view and (b) side view of the device.

TABLE I
TECHNICAL SPECIFICATIONS OF THE COMPONENTS

Raspberry Pi 4	CPU	Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit@1.5GHz
	Memory	4 GB
	Dimensions	88 x 58 x 19.5mm; 46g
Raspberry Pi Camera Module 2	Max image transfer rate	1080p: 30fps (encode and decode) 720p: 60fps
	Dimensions	23.86 x 25 x 9mm; 3g
YDLiDAR X4	Sensor range	up to 10m
	Resolution	~0.5 Deg
	Dimensions	68 x 58 x 101mm; 180g

wearing skirts were filmed from the side, using a CSI-2 extension cable. Videos were only used to associate each LiDAR scan with the position of the worker's legs (labeling). The technical specifications are in Table I.

III. DATA COLLECTION

A. Leg Positions to Recognize

The system recognizes six common leg positions: legs parallel with an angle of $\sim 90^\circ$ between upper and lower legs (of the six positions, this is the only correct position [17]), legs slightly open with knees bent, legs crossed, one leg on the other leg's knee, legs extended forward, and backward.

B. Device Positioning and Calibration

The LiDAR device was positioned in various ways while watching real-time scans. The best positioning was in front of the knees, 80–100-cm apart (based on the desk depth) as this positioning generated scans from which the leg position was easily understandable.

Calibration consisted of measuring height h of the participant's knees from the ground, and making him/her sit on an ergonomic chair with lower legs perpendicular to the floor. The chair height was adjusted so that the angle between the participant's upper and lower legs was $\sim 90^\circ$. The LiDAR sensor was then positioned in front of the participant, at height h from the floor, and slightly moved up and down while watching real-time scans: the more points outlining the legs in the scan, the better. Fig. 3 shows three scans generated during calibration. Blue ellipses were drawn to highlight the points that do not outline the legs in uncalibrated scans.

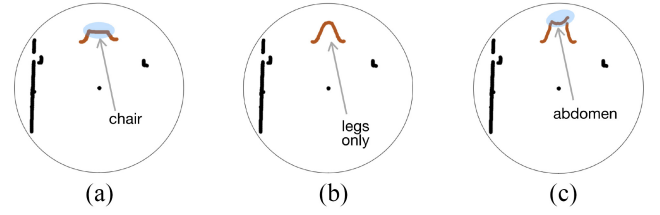


Fig. 3. LiDAR scans during calibration: (a) below and (c) above the knees and (b) at knee height. No blue-shaded area exists when the sensor is calibrated.

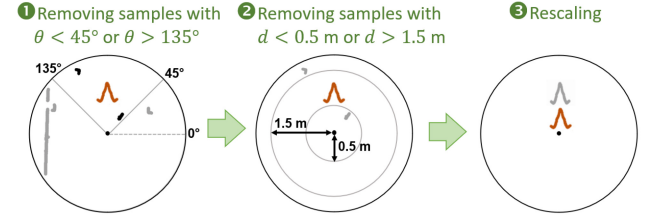


Fig. 4. Preprocessing steps performed on a sample obtained by one rotation of the LiDAR sensor. Red dots outline the legs and gray dots are those eliminated.

C. Dataset

A total of 30 healthy workers read and signed an informed consent, and voluntarily accepted to undergo continuous filming and LiDAR scanning of their legs. The informed consent also stated that data would be anonymized, and videos would be deleted after labeling LiDAR data.

Data were collected during normal activities (8 h) without any interaction between workers and researchers. Workers (50% women) were from 1.55 to 1.87-m tall, wearing trousers or skirts. Nearly half of the female participants (seven) wore a skirt. Fig. 5 shows two participants and their LiDAR scans. LiDAR scan frequency was set to 1 Hz, thus collecting 30 000 scans per worker. As videos showed that workers changed position every ~ 10 min (on average), scans were very similar to each other during these periods. Then, 10% of the scans of each position were randomly selected to get nuances of that position: the other scans were discarded. The number of scans per position ranged from 500 to 532. A balanced dataset was generated selecting 500 scans per position, for a total of 3000 samples (labeled scans).

IV. METHOD

A. Data Preprocessing

For each scan, the pairs (θ, d) with $d = 0$ were neglected, and those with $\theta < 45^\circ$ or $\theta > 135^\circ$ were removed to isolate the legs. Pairs with $d < 0.5$ or $d > 1.5$ were discarded as workers were sitting ~ 1 m from the sensor. Distances were rescaled as follows. For each scan $s = ((\theta_1^s, d_1^s), \dots, (\theta_n^s, d_n^s))$, the minimum distance d_{MIN}^s was determined. Among all d_{MIN}^s (one per scan), the minimum d_{MIN} was found. The rescaled distance r_i^s of each d_i^s was calculated as follows:

$$r_i^s = d_i^s - (d_{MIN}^s - d_{MIN}) \quad (1)$$

thus obtaining a *rescaled distance vector* $\mathbf{r}^s \in \mathbb{R}_+^{n^s}$ per sample, with n^s elements. Fig. 4 shows the preprocessing steps.

B. Feature Extraction and Data Partitioning

A total of 18 features were extracted from each $\mathbf{r}^s$: arithmetic, geometric and harmonic means, max, min, median,

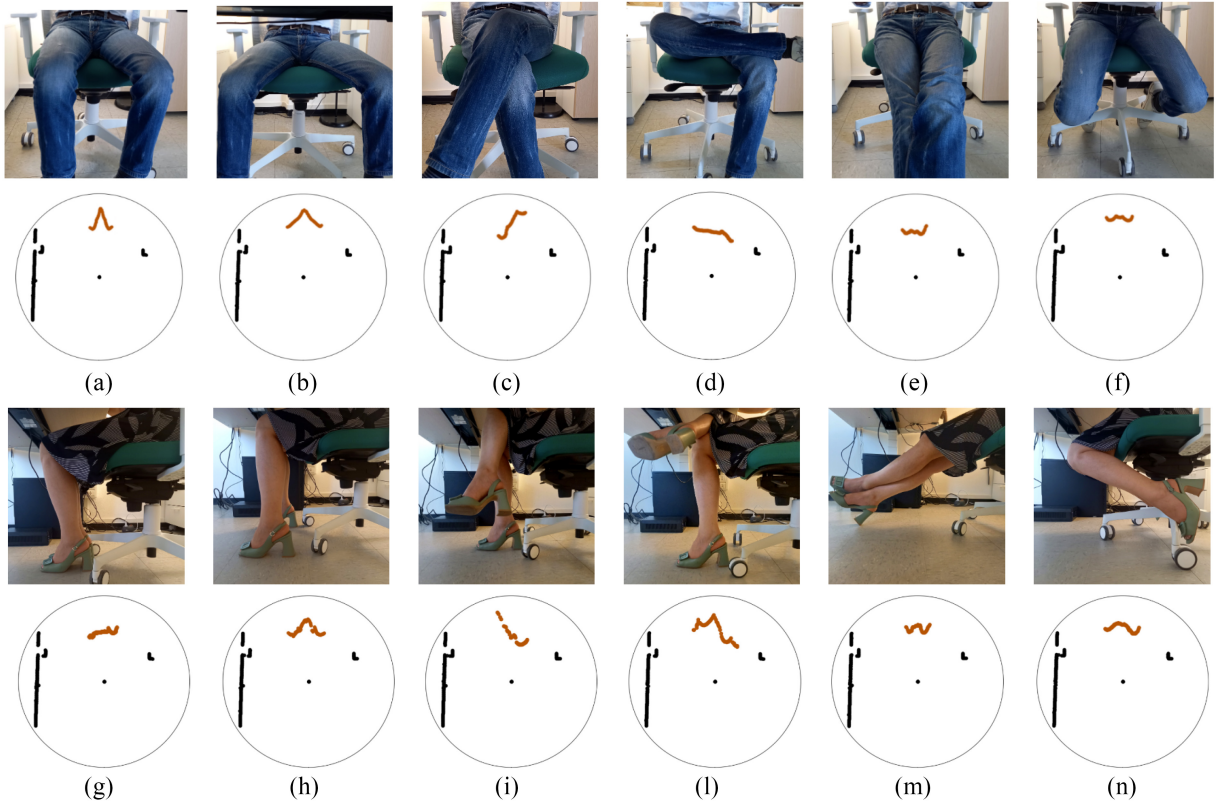


Fig. 5. (a)–(f) Front views of the legs of a worker wearing trousers and corresponding LiDAR scans for each position (1st and 2nd rows); and (g)–(n) side views of a worker wearing a skirt and corresponding LiDAR scans for each position (3rd and 4th rows). Red and black points outline the legs and furniture/walls, respectively.

mode, max-to-min difference, std deviation, variance, kurtosis, skewness, number of peaks, mean height of the peaks, std deviation of the heights of the peaks, ratio of the highest absolute value to the root-mean-square value, root-sum-of-squares level, and root-mean-square value. The dataset was split into subsets *A* and *B* ($A \cap B = \emptyset$), each containing a balanced number of samples per position. Subset *A* (80%) was used to find the best features from those extracted, and to train various types of classifier; subset *B* (20%) was used for testing.

C. Feature Selection

Features were selected via the sequential feature selection (SFS) [18], using accuracy (ratio of the number of samples correctly classified to the total) as a criterion. An SVM classifier was trained via *k*-fold cross-validation ($k=10$), with folds all having the same number of samples per class, and mean accuracy on the test set as a criterion. After 30 executions of the SFS, the most selected features were: kurtosis, skewness, harmonic mean, ratio of the highest absolute value to the root-mean-square value, and root-sum-of-squares level.

D. Architecture of the Classifier

The classifier was implemented as an SVM whose hyperparameters (type of kernel and value of γ) were optimized via grid search. One-versus-One and One-versus-All SVMs were tested. *Linear*, *quadratic*, and *Gaussian* kernels, and values of γ from 0 to 1, with step 0.01 were considered. For each combination of the hyperparameters, the SVM was trained using *k*-fold cross-validation with $k=10$. The performance

was measured using the mean accuracy on the test folds. One-versus-One SVMs with a Gaussian kernel and $\gamma = 0.56$ were the best.

E. Raspberry Application

The Raspberry application (developed in Python) waits for a LiDAR scan and then: 1) preprocesses the scan (see Section IV-A); 2) calculates the features (see Section IV-C); 3) classifies the scan using the SVM; and 4) stores the result in memory, along with the current timestamp.

V. RESULTS

A. Classification Accuracy of the SVM

The accuracy of the best SVM was 99.2%. Fig. 6a shows its confusion matrix on subset *B*. As can be seen, the SVM only confused position *b* (legs slightly open) with position *e* (legs extended forward) four times, and position *d* (one leg on the other leg's knee) with the correct position (*a*) once.

B. Comparison With Other Classifiers

The performance of the SVM was compared to those of other types of classifiers: *k*-NN, DT, and multilayer perceptron (MLP). After selecting various parameter setups, 30 models per type were trained for each setup. Fig. 6(b) shows the box plot of accuracy when using the best parameter setup for each type of classifier.

1) *k*-NN Classifiers: The *k*-NN was tested using equal and inverse weights, Euclidean and Manhattan distances, and $k \in$

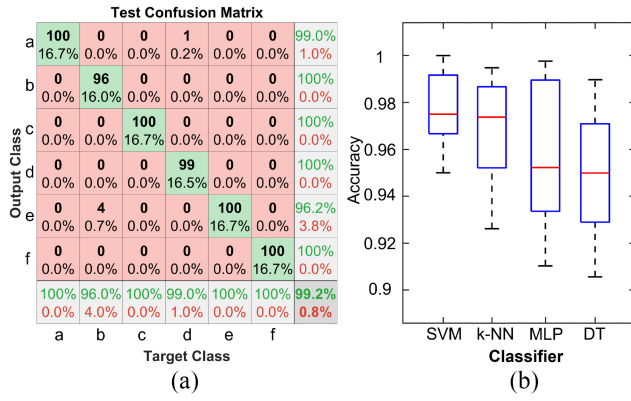


Fig. 6. (a) Confusion matrix of the best SVM on subset *B* and (b) box plots of accuracy of the SVM and compared classifiers.

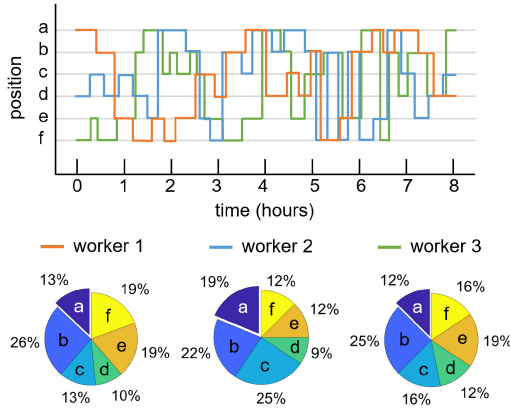


Fig. 7. Timeline of the positions of the three workers during 8 h (top), and corresponding pie charts (bottom).

$\{1, \dots, 6\}$ (higher values of k did not improve performance). The highest mean accuracy (97.4%) was achieved using inverse weights, Euclidean distance, and $k = 1$.

2) *MLP Classifiers*: MLPs were trained with one hidden layer and a number of neurons in $\{3, \dots, 15\}$, using the scaled conjugate gradient, Levenberg–Marquardt (L–M), and Bayesian regularization algorithms. Nine hidden neurons and the L–M algorithm led to the highest mean accuracy (95.8%). More neurons or hidden layers did not improve performance.

3) *DT Classifiers*: DTs were trained using the *Gini* criterion, with a max tree depth $D \in \{5, \dots, 15\}$, max number of splits S , and min number of leaf nodes L both in $\{3, \dots, 8\}$, and number of features $F \in \{1, \dots, 6\}$. The highest mean accuracy (95.3%) was achieved with $D = 6$, $S = 7$, $L = 3$, and $F = 4$.

C. Continuous Detection

Continuous detection was tested for 8 h involving three workers: two men (workers 1 and 3) and a woman (worker 2). For each worker, a timeline and a pie chart were generated to analyze the position changes over time, along with the percentage of time spent in each position, respectively.

For example, the timeline of worker 1 (top of Fig. 7) shows that he started the day with his legs in the correct position, after ~ 20 min, he opened the legs which were extended forward after ~ 15 min, with consequent arching of the back and increased spinal overload [17].

Pie charts (bottom of Fig. 7) show that workers maintained an incorrect position for 87%, 81%, and 88% of the day,

respectively, for a total of ~ 6 h 50', on average. The longest-lasting incorrect position of workers 1 and 3 was position *b* (legs slightly open with knees bent) which is not so dangerous. Instead, worker 2 spent 25% of the day with her legs crossed (position *c*), resulting in trunk misalignment and increased risk of back pain in the medium to long term [17].

VI. CONCLUSION

This letter has presented a discreet system that uses 2-D LiDAR and an SVM to recognize six leg positions while working at a desk with an accuracy of 99.2%. Without using body sensors or cameras, the system helps to determine the periods of the day a worker spends sitting with poor posture and how long these periods last. The risk for those who sit for long periods can thus be assessed without disturbing the worker but still respecting their privacy.

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